

Name:

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Score: _____

1. Prove: If H and K are subgroups of a group G , then $H \cap K$ is a subgroup of G .

Proof (Direct) Suppose H and K are subgroups of G . Let $M = H \cap K$.

The following considerations show M is a subgroup.

- ① Note $M \neq \emptyset$ because since $e \in H$ and $e \in K$, it follows $e \in H \cap K = M$.
- ② Let $a, b \in M$. We claim $ab^{-1} \in M$.
First, since $a \in M = H \cap K$, $a \in H$ and $a \in K$. For the same reason, $b \in H$ and $b \in K$. Then $ab^{-1} \in H$, as both $a \in H$ and $b^{-1} \in H$. Also $ab^{-1} \in K$ because $a, b^{-1} \in K$.
Consequently $ab^{-1} \in H \cap K = M$, as claimed.

By the second subgroup criterion, $H \cap K$ is a subgroup of G $\square$

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1. Suppose G is a group and $ab = a^{-1}b^{-1}$ for any two elements $a, b \in G$. Prove that G is abelian.

Proof (Direct)

Suppose $ab = a^{-1}b^{-1}$ for any $a, b \in G$.
Put $b = e$, and this becomes $ae = a^{-1}e^{-1}$
or $a = a^{-1}$ for every $a \in G$.

This means $aa = e$, or rather $\boxed{a^2 = e}$
for every element $a \in G$.

Now take any two $a, b \in G$. The
previous paragraph states $(ab)^2 = e$

Therefore: $(ab)(ab) = e$

$$abab = e$$

$$a abab = ae$$

$$ebab = a$$

$$bab = a$$

$$babb = ab$$

$$bae = ab$$

$$ba = ab.$$

Thus we've shown $ba = ab$ for
all $a, b \in G$, so G is abelian A small square with a diagonal line from the bottom-left to the top-right, used as a QED symbol to indicate the end of a proof.